Geometric Response of Exactly Degenerate Quantum State Bundles

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An exactly degenerate multiplet has no internal level spacings, but its projector can carry a non-Abelian quantum geometry over coupling space. We ask when this geometry admits an eigenstate-thermalization-type statistical closure. For independent centered complex response channels with deterministic weights, we derive exact finite-channel cumulant additivity, including ordinary covariance and pseudocovariance, and obtain a conditional fourth-cumulant law proportional to N_{eff}^{-1}. A separate response-algebra criterion identifies irreducibility but does not imply Gaussian statistics. We test the conditional law prospectively in a chiral rectangular-kernel ensemble with index-protected nullity $D = n_a - n_b$. The prediction, estimator, development band, three validation sizes, and control rule were hashed before validation. All exact-nullity, gap, response, gauge, and covariance gates pass. Nevertheless, the random-class values of $N_{\mathrm{eff}} R_4^{\mathrm{full}}$ are 1.29034, 1.36496, and 1.48145, all above the frozen band $[0.44763, 1.23521]$; local and repeated-cell controls separate still more strongly. Thus N_{eff} alone does not close the fourth-order response statistics in this registered ensemble. The test does not distinguish a failure of channel independence, standardized-cumulant comparability, or finite-size tangent closure. We formulate the resulting bounded geometric-response program for BPS sectors and black-hole-motivated models without claiming thermalization, universality, or a completed geometric ETH.

I. WHAT FAILS WHEN THE SPECTRUM IS EXACTLY DEGENERATE

The eigenstate thermalization hypothesis (ETH) is a statement about matrix elements of observables in an energy eigenbasis. In a microcanonical window, its usual form is

$$O_{mn} = \overline{O}(\overline{E})\delta_{mn} + e^{-S(\overline{E})/2} f_O(\overline{E},\omega) R_{mn}, \qquad \overline{E} = \frac{E_m + E_n}{2}, \qquad \omega = E_m - E_n. \tag{1}$$

Here O is a few-body operator, S is the thermodynamic entropy in the chosen symmetry sector, and R_{mn} is a fluctuating quantity with smooth local second moments. Equation (1) is useful because it relates an energy-basis statement to equilibrium expectation values and, with additional assumptions, to thermalization of isolated many-body systems [1–4]. Level statistics, spectral form factors, and operator matrix elements then give different, though not equivalent, views of the same nonintegrable dynamics.

An exactly degenerate protected sector removes the spectral coordinate on which Eq. (1) is organized. Let $H(\boldsymbol{\lambda})$ possess a rank-D eigenspace

$$H(\boldsymbol{\lambda})P(\boldsymbol{\lambda}) = E_0(\boldsymbol{\lambda})P(\boldsymbol{\lambda}), \qquad \operatorname{rank} P = D, \tag{2}$$

separated from its complement by a nonzero gap. For any two states in the range of P, one has $\omega = 0$. More importantly, the Hamiltonian does not select a basis within the fiber: an orthonormal frame $\Psi = (|\psi_1\rangle, \ldots, |\psi_D\rangle)$ may be replaced by ΨU for arbitrary $U \in U(D)$. Individual entries O_{mn} inside the degenerate block consequently depend on that choice. Coincident levels also make spacing ratios and the usual spectral form factor silent within the protected sector. This is a kinematic obstruction, not evidence for or against chaos.

Exact degeneracy occurs for several logically distinct reasons. Illustrative examples, discussed with their sources in Sec. V, include a symmetry-enforced finite multiplet, the common kernel of a frustration-free parent Hamiltonian, supersymmetric cohomology, commuting stabilizer constraints, and an index-protected rectangular-kernel null space. These mechanisms fix the energy but do not fix how the protected subspace changes when couplings are varied. Two Hamiltonian families can therefore have identical protected spectra and sharply different projector geometry. Conversely, a complicated projector response does not by itself imply conventional ETH, since no thermal observable or late-time state has yet entered the statement.

The natural variable is the family of projectors $P(\boldsymbol{\lambda})$ over a parameter manifold $\mathcal{M}$. Couplings λ^μ may be boundary twists, exactly marginal couplings, local Hamiltonian coefficients, or other controls that preserve the isolated multiplet. The derivative $\partial_\mu P$ measures motion of the subspace in the ambient Hilbert space. It is invariant under a change of frame within the fiber and remains defined when every protected energy is exactly equal. Adiabatic eigenstate deformation has also been used as a sensitive probe of ordinary quantum chaos away from exact degeneracy [5]; here the projector formulation is essential because the relevant object is a non-Abelian bundle.

This change of variable generalizes only the statistical question in ETH. It asks whether normalized, symmetry-resolved responses of a protected subspace obey a re-

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producible probability law. It does *not* replace energy conservation by parameter-space motion, and it does not turn a Berry curvature distribution into a proof of thermalization. In particular, the following implications are not assumed:

$$\begin{aligned} \text{random projector geometry} &\not\Rightarrow \text{thermalization}, \\ \text{exact degeneracy} &\not\Rightarrow \text{integrability}, \end{aligned} \tag{3}$$

and a scalar response-algebra commutant will not be treated as a sufficient condition for either Gaussian statistics or ETH.

Our use of the phrase *Geometric ETH* is therefore conditional. It denotes a proposed hierarchy of statistical closure statements for gauge-covariant response variables, defined in Sec. III. One finite-channel identity within that hierarchy is exact and is proved in Sec. IV. The random-channel closure required to promote that identity to a physical ensemble statement was tested prospectively in a chiral index model and did not pass its sealed validation criterion. The result of the present work is thus a theorem with explicit hypotheses and a falsified model-level extrapolation, not an establishment of Geometric ETH.

II. GEOMETRY OF AN ISOLATED DEGENERATE SUBSPACE

We now fix the geometric objects whose statistics can be compared without choosing a basis in the protected multiplet. Let $\mathcal{H}$ be a finite Hilbert space and let $H(\boldsymbol{\lambda})$ be smooth on a coordinate patch of $\mathcal{M}$. We assume that Eq. (2) holds with constant rank D and that

$$\Delta(\boldsymbol{\lambda}) = \operatorname{dist}\big(E_0(\boldsymbol{\lambda}), \operatorname{spec}\big[QH(\boldsymbol{\lambda})Q\big]\big) > 0, \qquad Q = 1-P, \tag{4}$$

throughout the patch. The fibers $P(\boldsymbol{\lambda})\mathcal{H}$ form a rank-D complex vector bundle. A local frame Ψ obeys $\Psi^\dagger\Psi = I_D$ and $P = \Psi\Psi^\dagger$. On an overlap of patches it transforms as

$$\Psi(\boldsymbol{\lambda}) \longmapsto \Psi(\boldsymbol{\lambda})U(\boldsymbol{\lambda}), \qquad U(\boldsymbol{\lambda}) \in U(D). \tag{5}$$

This is a change of coordinates in the same physical subspace.

The Wilczek–Zee connection and curvature in this convention are

$$\begin{aligned} A_\mu &= i\Psi^\dagger\partial_\mu\Psi, \\ F_{\mu\nu} &= \partial_\mu A_\nu - \partial_\nu A_\mu - i[A_\mu, A_\nu], \end{aligned} \tag{6}$$

which transform as a connection and by conjugation, respectively. Their Abelian rank-one antecedent and non-Abelian extension are standard consequences of adiabatic transport [6–8]. Connection entries at one point are gauge dependent; curvature eigenvalues, invariant traces, and Wilson-loop conjugacy classes are not.

The response leaving the fiber is

$$X_\mu = Q\partial_\mu\Psi \in \operatorname{Hom}(P\mathcal{H}, Q\mathcal{H}). \tag{7}$$

It transforms as $X_\mu \mapsto X_\mu U$. Differentiating the eigenvalue equation and projecting with Q gives the reduced-resolvent form

$$X_\mu = -R_Q(\partial_\mu H)\Psi, \qquad R_Q = Q(H - E_0)^{-1}Q. \tag{8}$$

The inverse in Eq. (8) is restricted to $Q\mathcal{H}$. Thus a closing external gap is a singularity of the construction rather than an unusually large sample from a fixed response ensemble.

The matrix-valued quantum geometric tensor (QGT) is the fiber endomorphism

$$\mathcal{Q}_{\mu\nu} = (\partial_\mu\Psi)^\dagger Q(\partial_\nu\Psi) = X_\mu^\dagger X_\nu. \tag{9}$$

Its Hermitian symmetric and antisymmetric parts give the quantum metric and Berry curvature,

$$\begin{aligned} G_{\mu\nu} &= \frac{1}{2}\left(\mathcal{Q}_{\mu\nu} + \mathcal{Q}_{\nu\mu}\right), \\ F_{\mu\nu} &= i\left(\mathcal{Q}_{\mu\nu} - \mathcal{Q}_{\nu\mu}\right). \end{aligned} \tag{10}$$

For $D = 1$, Eq. (10) reduces to $g_{\mu\nu} = \operatorname{Re}\mathcal{Q}_{\mu\nu}$ and $F_{\mu\nu} = -2\operatorname{Im}\mathcal{Q}_{\mu\nu}$, with the stated connection convention [9, 10]. For $D > 1$, each block transforms as $\mathcal{Q}_{\mu\nu} \mapsto U^\dagger\mathcal{Q}_{\mu\nu}U$.

It is useful to separate the central and traceless curvature,

$$\begin{aligned} F_{\mu\nu} &= F^{(0)}_{\mu\nu} I_D + \widetilde{F}_{\mu\nu}, \\ F^{(0)}_{\mu\nu} &= \frac{1}{D}\operatorname{Tr}F_{\mu\nu}, \qquad \operatorname{Tr}\widetilde{F}_{\mu\nu} = 0. \end{aligned} \tag{11}$$

The trace controls the determinant line bundle and, on a closed two-cycle $\Sigma \subset \mathcal{M}$, the first Chern number

$$C_1[\Sigma] = \frac{1}{2\pi}\int_\Sigma \operatorname{Tr}F. \tag{12}$$

This normalization follows the many-body Berry-curvature convention of Ref. [11]. The traceless part records internal $SU(D)$ rotation not fixed by C_1. Keeping these pieces separate prevents a topological mean curvature from being mistaken for fluctuating non-Abelian geometry.

Local statistical statements must be made from conjugation invariants or from covariantly compared matrices. At one point these include $D^{-1}\operatorname{Tr}\widetilde{F}^k$, the spectrum of $\widetilde{F}$, and invariant contractions of $\mathcal{Q}$. At different points, matrices must first be parallel transported to a common fiber, or the comparison must be restricted to scalar invariants. Raw entries of A_μ, $F_{\mu\nu}$, or X_μ in an arbitrary numerical frame are not primary observables.

The geometry also separates protection from mixing. The rank D and the gap in Eq. (4) state that a multiplet exists. The span of the responses X_μ, the algebra generated by $X_\mu^\dagger X_\nu$, and their higher connected moments state how extensively accessible deformations move that multiplet. The first set of facts can be exact while the second remains structured, reducible, or random. This separation is the starting point for the definitions below.

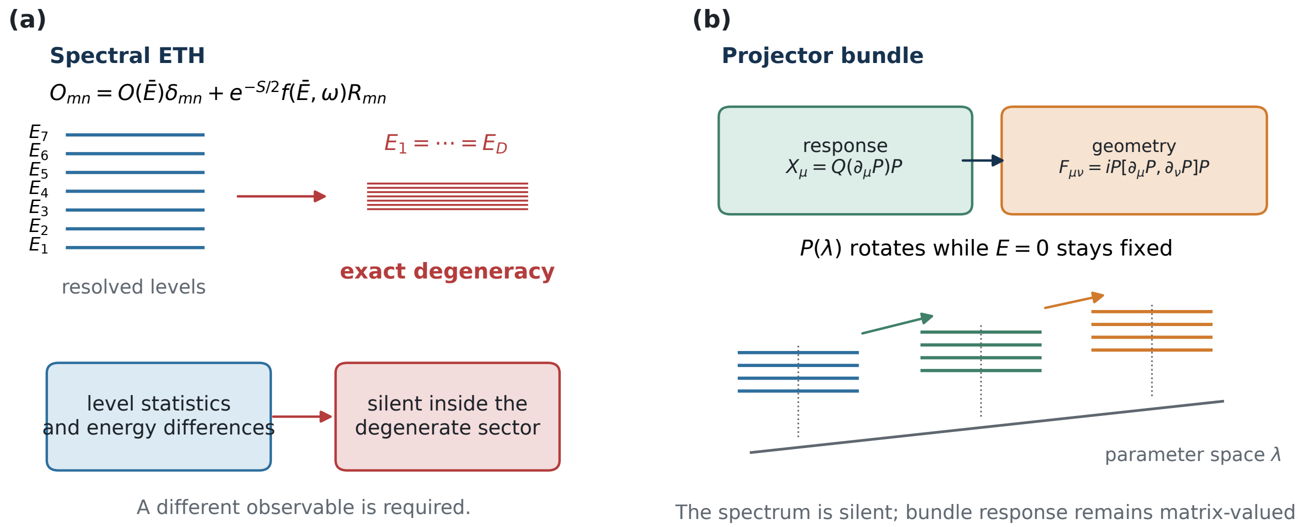

FIG. 1. From spectral ETH to a degenerate state bundle. Conventional ETH resolves energy eigenstates and organizes matrix elements by mean energy and energy difference. Exact degeneracy removes that internal spectral coordinate. The replacement considered here is the off-fiber response $X_a = -R_Q(\partial_a H)P$, whose bilinears give the non-Abelian quantum geometric tensor. The rightmost column separates exact definitions, statistical assumptions, and quantities that remain unproved.

III. A HIERARCHY OF GEOMETRIC STATISTICAL STATEMENTS

The bundle construction specifies observables but not a probability law. A statistical statement requires an ensemble and a limiting procedure. We therefore consider a sequence indexed by L, where L may denote particle number, volume, fiber rank, or another explicitly stated control. Within each L, all exact symmetries and invariant blocks are resolved before samples are pooled. A sample may be a disorder realization, a base point in a stationary parameter ensemble, or a Hamiltonian drawn from a specified population. These choices are not interchangeable and must be recorded as part of the definition.

Let a label an accessible tangent direction and write $x_a = \mathrm{vec}(X_a)$ in a local frame. We subtract the population mean and apply a deterministic normalization chosen without using the higher-order outcomes under test:

$$z_a = W_L\left(x_a - \mathbb{E}x_a\right). \tag{13}$$

Here W_L may remove a scalar variance or whiten a fixed symmetry-resolved support. A frame change acts linearly on z_a; the tensors below transform covariantly, while the final statements are made for their invariant contractions. Outcome-dependent whitening, post-selection of tangents, and refitting W_L after inspecting fourth-order data are excluded.

Complex response variables require two second-moment tensors. Suppressing the tangent label, define

$$C_{ij} = \mathbb{E}[z_i z_j^*], \qquad \Pi_{ij} = \mathbb{E}[z_i z_j]. \tag{14}$$

We use Π for pseudocovariance in the manuscript to avoid collision with the spectral projector P. A proper complex population has $\Pi = 0$, but properness must be checked rather than imposed. For a centered complex Gaussian vector the representative fourth moment is

$$\mathbb{E}[z_i z_j^* z_k z_l^*] = C_{ij}C_{kl} + C_{il}C_{kj} + \Pi_{ik}\Pi_{jl}^*. \tag{15}$$

The right-hand side contains all three Wick pairings. Subtracting only the two covariance pairings defines an incomplete null whenever $\Pi \neq 0$. Equivalently, one may realify z into $(\mathrm{Re}\, z, \mathrm{Im}\, z)$ and use its full real covariance.

We next define three levels of a possible Geometric-ETH statement. They are definitions of claims, not results of the present calculation.

a. Weak geometric closure. A sequence has weak geometric closure for a stated set of tangents and invariant observables if: (i) the centered, normalized two-point tensors (C_L, Π_L) have deterministic limits on their resolved support; and (ii) the selected quadratic gauge invariants, such as $D^{-1}\,\mathrm{Tr}\,\widetilde{F}^2_{\mu\nu}$ or contractions of $\mathcal{Q}_{\mu\nu}$, concentrate around the values determined by those limits. This definition makes no Gaussian assertion and places no condition on connected cumulants of order four and above. It is weaker than ordinary ETH because it contains neither the smooth diagonal function $\overline{O}(E)$ nor a thermalization statement.

b. Strong geometric closure. A sequence has strong geometric closure if it has weak closure and, after complete whitening on every symmetry-resolved support, every fixed finite-dimensional real-linear projection has a

Gaussian limit:

$$\kappa_k(f_1(z_L),\dots,f_k(z_L)) \longrightarrow 0, \qquad k \geq 3, \tag{16}$$

for all fixed test functionals for which the required moments exist. The limiting complex Gaussian is specified by both C and Π. We additionally require the accessible response algebra to have no unresolved invariant block in the fiber. This irreducibility condition is necessary for a full-matrix law, but it does not imply Eq. (16). Strong geometric closure remains a statement about parameter-space response, not about real-time thermalization.

c. Deformed geometric closure. A sequence has deformed geometric closure if the complete second moments stabilize but specified connected tensors retain a reproducible structure after whitening. Examples include a fixed improper-complex component, channel-dependent fourth cumulants, locality blocks, unequal gap weights in Eq. (8), or a nonzero limiting connected tensor. The deformation must be stated before comparison and carried by measured tensors or a specified probability law. Calling any non-Gaussian residual "deformed" without such a prediction would be tautological. A reducible symmetry mixture is also not a deformation; the blocks must first be separated.

These definitions distinguish covariance closure, Gaussian closure, and structured non-Gaussian closure. The distinction matters because a Wishart-like QGT can arise whenever X_a is approximately Gaussian, whereas curvature is a difference of correlated bilinears and can preserve memory of Π, unequal channel weights, or local constraints. Agreement of an eigenvalue histogram with a familiar random-matrix curve tests only the particular marginal shown. Strong closure requires the complete covariance and higher connected tensors, not a single spectral statistic.

The model used for the exact result below is more restrictive than any of these definitions. It assumes that a normalized response can be decomposed as

$$Z = \sum_{\alpha=1}^{n} w_\alpha Y_\alpha, \qquad \mathbb{E}Y_\alpha = 0, \tag{17}$$

where the Y_α are independent complex random matrices in one common response space and the real weights w_α are deterministic. We call Eq. (17) the *independent-channel model*. It is not a consequence of exact degeneracy, a spectral gap, or an irreducible response algebra. Establishing it for a physical family requires identifying the independent units and showing that mixed connected cumulants are absent or controlled.

The distinction between definition and model fixes the logic of the paper. Section IV derives an exact property of Eq. (17). The numerical oracle checks the implementation on synthetic channels drawn from the assumed population. The chiral calculation then asks prospectively whether a new protected model obeys the resulting fixed prediction. It does not: the random tangent class misses the sealed validation band. Consequently neither weak, strong, nor deformed geometric closure is established here for a general physical model class. The failed test instead identifies independence and channel stationarity as substantive hypotheses rather than consequences of a large degenerate subspace.

For later reference, we reserve *Geometric ETH* for the conjecture that an explicitly specified family has at least weak closure and, after all protection-induced structures are accounted for, approaches either the strong or a preregistered deformed class. This conjecture can be true for one protection and tangent ensemble and false for another. No asymptotic, universal, gravitational, or thermalization form of the conjecture follows from the finite-channel theorem.

IV. INDEPENDENT CHANNELS: EXACT STATEMENT AND LIMITED CONSEQUENCE

We now isolate the part of the analysis that is a theorem. Let

$$\mathcal{V} = \mathrm{Hom}(P\mathcal{H}, Q\mathcal{H}) \tag{18}$$

at a fixed base point. Cumulants of a complex matrix in $\mathcal{V}$ are defined on the underlying real vector space; complex trace words are obtained by complexifying the resulting real multilinear tensors. This convention retains holomorphic, antiholomorphic, and mixed contractions and is equivalent to retaining both tensors in Eq. (14).

a. Theorem 1 (weighted cumulant additivity). Let $Y_1,\dots,Y_n \in \mathcal{V}$ be mutually independent centered complex random matrices, and let $w_1,\dots,w_n \in \mathbb{R}$ be deterministic. For an order-k statement, assume that every real-linear projection under consideration has a finite kth absolute moment. All channels must occupy the same response space, or be identified with it by a fixed map independent of the outcome. Then, for Z in Eq. (17),

$$\boxed{\kappa_k(Z) = \sum_{\alpha=1}^{n} w_\alpha^k \kappa_k(Y_\alpha)}. \tag{19}$$

The equality is a finite-n tensor identity. It is not a central-limit approximation.

To see the result, let t be any real-linear test field on $\mathcal{V}$. Independence factorizes the characteristic function and hence its logarithm,

$$\log \mathbb{E}e^{it(Z)} = \sum_{\alpha=1}^{n} \log \mathbb{E}e^{iw_\alpha t(Y_\alpha)}. \tag{20}$$

Taking k derivatives at the origin supplies w_α^k and produces no mixed-channel connected term. Polarization over distinct test fields proves Eq. (19). A detailed realification proof is given in the Supplemental Material.

At second order, with $y_\alpha = \mathrm{vec}(Y_\alpha)$, the ordinary covariance and pseudocovariance are

$$C_\alpha = \mathbb{E}[y_\alpha y_\alpha^\dagger], \qquad \Pi_\alpha = \mathbb{E}[y_\alpha y_\alpha^T], \tag{21}$$

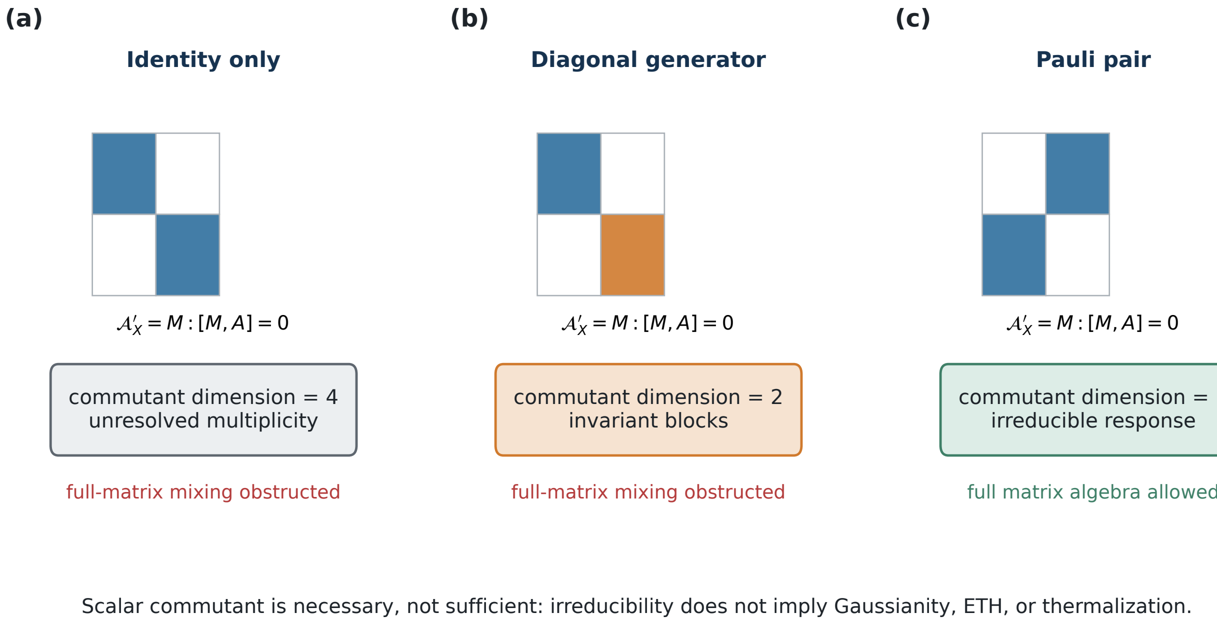

FIG. 2. Accessible response algebras. The generators $X_a^\dagger X_b$ can be frozen, central, reducible, or irreducible on the protected fiber. A nonscalar commutant identifies invariant blocks and forbids an unresolved full-matrix ansatz. A scalar commutant removes that obstruction but does not imply Gaussianity, cumulant decay, thermalization, or ETH.

and independence gives

$$C_Z = \sum_\alpha w_\alpha^2 C_\alpha, \qquad \Pi_Z = \sum_\alpha w_\alpha^2 \Pi_\alpha. \tag{22}$$

The connected fourth tensor must subtract the Wick tensor constructed from both C_Z and Π_Z. Pairwise uncorrelated channels are insufficient for Theorem 1: mixed fourth cumulants can remain when independence fails.

The participation number of the deterministic weights is

$$N_{\text{eff}} = \frac{\left(\sum_\alpha w_\alpha^2\right)^2}{\sum_\alpha w_\alpha^4}. \tag{23}$$

For L^2-normalized equal weights it equals n. Concentrated weights can make it much smaller than the nominal channel count. Theorem 1 alone does not produce an inverse-N_{eff} law because the tensors $\kappa_4(Y_\alpha)$ need not be equal or even comparable.

b. Corollary 1 (conditional fourth-cumulant law). Assume in addition that the tested channel contraction is variance normalized, every channel has a finite fourth moment, and the standardized fourth cumulants are identical, $\kappa_4(Y_\alpha) = K_4$. Then

$$\frac{\kappa_4(Z)}{\left(\sum_\alpha w_\alpha^2\right)^2} = \frac{K_4}{N_{\text{eff}}}. \tag{24}$$

If instead $\|\kappa_4(Y_\alpha)\| \le K$ in one fixed tensor norm, then

$$\frac{\|\kappa_4(Z)\|}{\left(\sum_\alpha w_\alpha^2\right)^2} \le \frac{K}{N_{\text{eff}}}. \tag{25}$$

For unequal known channel cumulants, the executable prediction is

$$\kappa_{4,\text{pred}}(Z) = \frac{\sum_\alpha w_\alpha^4 \kappa_4(Y_\alpha)}{\left(\sum_\alpha w_\alpha^2\right)^2}. \tag{26}$$

No exponent is fitted in Eqs. (24)– (26).

The analytic implementation was checked with a fixed non-Gaussian complex matrix population having zero mean, unit variance, zero population pseudocovariance, finite fourth moment, and standardized fourth cumulant $K_4 = 8$. Equal and linearly unequal weights were evaluated at $n = 4, 8, 16, 32, 64$ using 50,000 matrices and 25 uncertainty blocks. The largest direct relative error from Eq. (26) was 0.04434, below the registered 0.08 implementation threshold. This oracle verifies the code path under data generated from the theorem assumptions. It is not evidence that the response channels of a physical Hamiltonian are independent.

The second exact statement concerns only accessibility of the fiber. Given response matrices X_a, define the unital star-closed algebra

$$\mathcal{A}_X = \text{alg}^*\{X_a^\dagger X_b : a, b\} \subseteq \text{End}_{\mathbb{C}}(P\mathcal{H}) \tag{27}$$

and its commutant

$$\mathcal{A}'_X = \{M : [M, A] = 0 \text{ for every } A \in \mathcal{A}_X\}. \tag{28}$$

c. Proposition 1 (commutant criterion). For a finite-dimensional complex star algebra, $\mathcal{A}'_X = \mathbb{C} I_D$ implies that

$\mathcal{A}_X$ acts irreducibly on the protected fiber. Burnside's theorem then gives

$$\mathcal{A}_X = \mathrm{End}_{\mathbb{C}}(P\mathcal{H}). \tag{29}$$

A nonscalar commutant therefore rules out an unrestricted full-matrix mixing ansatz until its invariant blocks are resolved. The converse use relevant to statistics is forbidden: a scalar commutant does not imply independent channels, Gaussianity, suppression of connected cumulants, locality, thermalization, or ETH. A deterministic pair of matrices can generate the full matrix algebra.

The theorem and proposition leave several explicit failure branches. Mixed connected tensors modify Eq. (19) for correlated channels. Without finite fourth moments, the statistic in Eq. (24) is undefined. If $\max_\alpha w_\alpha^2/\sum_\beta w_\beta^2$ does not decrease, nominal channel proliferation does not imply growing N_{eff}. Unresolved symmetry blocks produce a nonscalar commutant; nonstationary tangent populations invalidate a comparison based only on N_{eff}; and a closing external gap invalidates the isolated-projector response. Finally, omitting pseudocovariance removes the third Wick pairing and changes the null being tested.

These limitations are active in the result boundary. The analytic audit and exact index construction passed, but the prospectively sealed chiral random tangent validation did not remain inside its fixed channel-law band. The selected branch is therefore `random_channel_failure`. Retrospective data from other protected models lack documented independent response units and cannot repair that prospective failure. We retain Eqs. (19)–(29) as exact conditional statements and reject the stronger inference that they establish Geometric ETH for exactly degenerate quantum-state bundles.

V. PROTECTION CLASSES AND RESPONSE ALGEBRAS

Exact degeneracy is a kinematic condition, not a probability distribution. A useful classification must therefore keep two questions separate: what enforces the multiplicity of the protected fiber, and what algebra is generated when the fiber is moved. Table I gives six illustrative mechanisms organized by this distinction; it is not an exhaustive literature classification. The entries are not universality classes. In particular, two Hamiltonians with the same nullity formula can have different commutants, and two response families with scalar commutants can have different fourth cumulants.

For a common-kernel parent Hamiltonian, positivity and frustration freedom give

$$H(\lambda) = \sum_j A_j(\lambda)^\dagger A_j(\lambda), \qquad \mathcal{E}_\lambda = \bigcap_j \ker A_j(\lambda). \tag{30}$$

The dimension of $\mathcal{E}_\lambda$ can remain constant even when its embedding in the ambient Hilbert space changes. Fractional-quantum-Hall clustering states and the Kapit–Mueller lattice parent provide common-kernel examples [12–14]. In the response problem considered here, a boundary twist is treated as a deformation of the constraint family and may move that kernel. The parent construction does not say whether $X_a^\dagger X_b$ is central, block diagonal, or irreducible.

In a cohomological system,

$$H = \{Q, Q^\dagger\}, \qquad Q^2 = 0, \tag{31}$$

and the zero modes represent cohomology classes. Lattice $\mathcal{N} = 2$ models supply a concrete many-body realization [15, 16]; supersymmetric SYK supplies a random-coupling protected model [17]. The exact and coexact complements need not contribute equally to the response. Their imbalance is visible as a nonscalar commutant or as Hodge-resolved covariance, even when the total zero-mode rank is large.

Commuting-projector stabilizer Hamiltonians represent the opposite extreme. Positive coefficient changes can leave every stabilizer eigenspace fixed, so $\partial_a P = 0$ and all geometric observables vanish. An isospectral conjugation can move the same code space while producing a central curvature. The X-cube code supplies a three-dimensional subsystem example with size-dependent exact degeneracy [18]. These solvable paths show why a moving projector and nonzero curvature are not sufficient conditions for stochastic non-Abelian geometry.

Symmetry and integrability protect multiplets in a different way. Before any statistical statement is made, the fiber must be resolved into irreducible symmetry or charge sectors. Otherwise the commutant contains the unresolved projectors by construction. The criterion $\mathcal{A}_X' = \mathbb{C}I_D$ is consequently a statement about an already resolved fiber. It eliminates one algebraic obstruction to full-matrix mixing but does not establish Gaussian closure.

The chiral/index class is especially useful for a prospective test because its nullity is fixed by a rectangular matrix index, while the tangent distribution can be specified independently. It is neither a fractional-quantum-Hall parent nor a supersymmetric cohomology model. We now use this class to test whether the effective-channel variable derived in Sec. IV closes the fourth cumulant of a new exactly degenerate ensemble.

The finite-size calculations behind the condensed-matter entries in this comparison are reported in the companion Paper I [19]. The present paper imports its frozen evidence registry only for mechanism labels and bounded cross-model context; it does not import the companion prose or treat those retrospective cases as validation data for the chiral prediction.

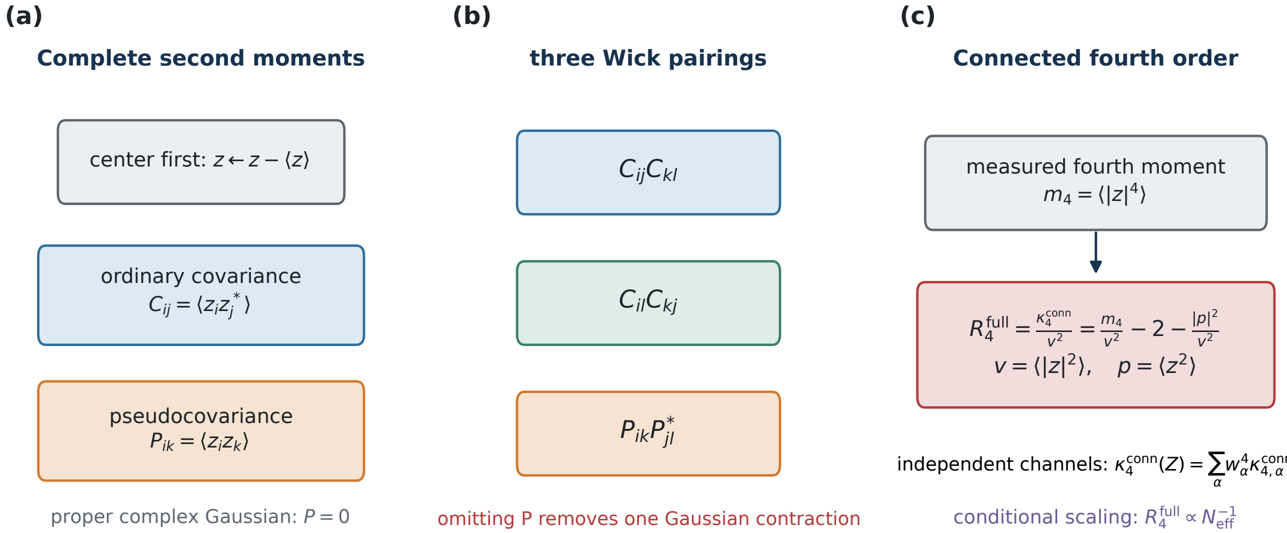

FIG. 3. Hierarchy of complex statistical closure. Centering is followed by the ordinary covariance C and pseudocovariance Π; together they determine all three Wick pairings. The connected fourth tensor is the residual after those pairings. For independent weighted channels it is additive exactly. The N_{eff}^{-1} specialization additionally assumes comparable standardized single-channel cumulants; the chiral calculation rejects the combined finite-size specialization without isolating which assumption fails.

VI. A SEALED CHIRAL-INDEX TEST

A. Exact kernel and horizontal response

Let $T \in \mathbb{C}^{n_b \times n_a}$ have full row rank, with $n_a > n_b$, and define

$$H_T = \begin{pmatrix} 0 & T^\dagger \\ T & 0 \end{pmatrix}, \qquad \Gamma = \begin{pmatrix} I_{n_a} & 0 \\ 0 & -I_{n_b} \end{pmatrix}. \tag{32}$$

The anticommutator $\{H_T, \Gamma\}$ vanishes. The index fixes an exactly degenerate zero-energy fiber of rank

$$D = \dim \ker T = n_a - n_b, \tag{33}$$

and the external gap is the smallest singular value $\sigma_{\min}(T)$. If K is an $n_a \times D$ orthonormal frame for $\ker T$, differentiating $TK = 0$ along a tangent $\dot{T}$ and imposing the horizontal gauge $K^\dagger \dot{K} = 0$ gives

$$\dot{K} = -T^\dagger (TT^\dagger)^{-1} \dot{T} K. \tag{34}$$

Equation (34) is the exact off-fiber response. It uses only the rectangular factor and its reduced singular system; the dense $(n_a + n_b)$-dimensional Hamiltonian is unnecessary in production.

For every base matrix we verify the index, kernel residual, frame orthonormality, positive gap, horizontal-gauge equation, centered finite difference of the projector, and covariance under independent unitary changes of the two sublattice bases. Across 576 registered base/class records the gap lies between 0.1570 and 0.2570. The largest relative projector finite-difference error is 5.16e-09, the largest gauge-covariance error is 7.84e-15, and the largest whitening identity error is 1.71e-14.

B. Tangent populations and effective channels

The base factor has independent proper-complex Gaussian entries with variance scaled by n_a^{-1}. We compare three tangent populations at equal Frobenius norm and equal channel count. The *random* population has independent dense proper-complex entries. The *local* population is supported in a fixed-width band around the map from the n_b rows to the n_a columns. The *structured* population repeats one 4×6 complex cell across the rectangular factor. Base and tangent seed streams are disjoint across the three populations.

Writing $T = U\Sigma V^\dagger$, the response strength carried by singular row i for a tangent a is

$$q_{ia} = \sum_{\mu=1}^{D} \frac{|(U^\dagger \dot{T}_a K)_{i\mu}|^2}{\sigma_i^2}, \qquad N_{\mathrm{eff},a} = \frac{(\sum_i q_{ia})^2}{\sum_i q_{ia}^2}. \tag{35}$$

This participation ratio is fixed before the fourth moment is evaluated. At each base, the 16 tangent responses are

TABLE I. Illustrative, non-exhaustive mechanisms that produce an exactly degenerate subspace. The protection law determines the existence and dimension of the fiber, whereas the accessible tangent set determines the response algebra. Neither column by itself fixes a random-matrix ensemble.

Protection class	**Exact degeneracy**	**Motion of P**	**Response-algebra obstruction**	**How exactness is lost**
Common kernel / parent	Positive constraints share a null space; quasihole or clustering rules fix D.	Constraint tensors or boundary twists move the common kernel.	Tangent choice may be central, reducible, or full; the parent form does not determine statistics.	A generic term not annihilating the kernel, Landau-level mixing, or a gap closing.
Cohomological	$H = \{Q, Q^\dagger\}$ and $Q^2 = 0$ identify zero modes with cohomology.	Deformations of Q move exact and coexact complements.	Hodge blocks can leave a nonscalar commutant even when curvature is nonzero.	Breaking nilpotency or supersymmetry, or changing cohomology across a singular point.
Stabilizer	Mutually commuting projectors and logical operators fix an exact code space.	Coefficient changes leave P fixed; conjugations can move it isospectrally.	Frozen or scalar response is common at solvable points.	Noncommuting perturbations, boundaries, or virtual logical processes at finite size.
Chiral / index	Full-row-rank $T \in \mathbb{C}^{n_b \times n_a}$ gives $D = n_a - n_b$.	Rectangular-factor tangents move $\ker T$ while the index stays fixed.	Random tangents need not close under N_{eff} alone, as the sealed test shows.	Rank loss, a closing singular-value gap, or chiral-symmetry breaking.
Symmetry multiplet	An irreducible representation, Kramers theorem, or conserved charge fixes a multiplet.	Symmetry-preserving couplings rotate the representation bundle.	The commutant contains unresolved symmetry blocks until sectors are resolved.	A field or perturbation that breaks the protecting symmetry.
Integrable / spectrum-generating algebra	Hidden commuting charges or a spectrum-generating algebra enforce multiplets.	Algebra-preserving parameter changes transport the representation.	A large commutant is expected and obstructs unrestricted mixing.	Generic charge-breaking terms, truncation of long-range couplings, or disorder.

centered over their matrix entries and transformed by the same registered tangent-channel covariance-whitening rule. For each resulting complex scalar sample z, define $\mu_z = \langle z \rangle$, $\delta z = z - \mu_z$, and $v = \langle |\delta z|^2 \rangle$. We subtract the complete Gaussian Wick tensor and report the dimensionless statistic

$$R_4^{\text{full}} = \frac{\kappa_4^{\text{conn}}}{v^2} = \frac{\langle |\delta z|^4 \rangle}{v^2} - 2 - \frac{|\langle (\delta z)^2 \rangle|^2}{v^2}. \tag{36}$$

The pseudocovariance term in Eq. (36) is essential: an improper real Gaussian sample viewed as complex has zero complete cumulant but would give a spurious residual under a proper-complex two-pairing subtraction.

C. Prediction seal

The complete development grid consists of $(16, 24)$, $(24, 36)$, and $(32, 48)$, with 64 independent bases and 16 tangents per base and class. It was opened before the validation calculation and used only to form the simultaneous random-class band

$$N_{\text{eff}} R_4^{\text{full}} \in [0.44763, 1.23521]. \tag{37}$$

The prediction file, its source hashes, the statistic, the three validation sizes $(48, 72)$, $(64, 96)$, and $(96, 144)$, the estimator, and the branch precedence were then written and hashed before any validation outcome existed. The primary decision uses the point estimate obtained by averaging the 16 tangent statistics within each base and then averaging the 64 base means. All three random-class validation estimates must lie inside Eq. (37). Ten thousand base-bootstrap replicates provide Bonferroni simultaneous intervals for reporting only; they do not widen the sealed band.

A control is declared separated only if the same local or structured class has a bootstrap interval disjoint from the band at at least two of the three validation sizes, including $(n_b, n_a) = (96, 144)$. The four branches, in precedence order, are feasibility failure, random-channel failure, deformed locality, and channel-law validation. Thus exactness cannot be rescued by a favorable statistic, and a failed random prediction cannot be reclassified after inspecting the controls.

D. Validation result

Figure 4 shows the sealed result. The random-class estimates are

$$\begin{aligned} (48, 72) &: \quad 1.29034[1.20517, 1.37573], \\ (64, 96) &: \quad 1.36496[1.27338, 1.46219], \\ (96, 144) &: \quad 1.48145[1.39258, 1.56772]. \end{aligned} \tag{38}$$

Every point lies above the frozen upper boundary 1.23521. The registered random-channel prediction therefore fails at all three validation sizes, selecting `random_channel_failure`. The local estimates, 1.50856, 1.52806, and 1.80097, also lie above the band. The repeated-cell estimates, 8.75820, 11.62585, and 17.49123, are much larger. Both controls satisfy the predeclared separation rule, but control separation cannot overturn the failed primary prediction.

The failure does not contradict weighted cumulant additivity. That theorem predicts

$$\kappa_4^{\mathrm{conn}}(Z) = \sum_i w_i^4 \kappa_{4,i}^{\mathrm{conn}}, \tag{39}$$

whereas the reduction to a single constant divided by N_{eff} additionally requires channel independence and comparable standardized single-channel cumulants. Equation (35) retains the variance weights q_i but not possible dependence of $\kappa_{4,i}^{\mathrm{conn}}$ on the singular values, the kernel frame, or the tangent support. The validation establishes only that N_{eff} is insufficient for this registered finite-size ensemble. It does not distinguish correlated channels, noncomparable standardized channel cumulants, finite-rank geometry, or tangent-specific effects. Any amended closure law must preregister an additional observable that separates these possibilities. We do not fit such an observable to the opened validation data.

VII. BPS SECTORS, BLACK-HOLE DIAGNOSTICS, AND PROTECTED GEOMETRY

The motivation for a geometric formulation of eigenstate thermalization is particularly sharp in a supersymmetric sector. Ordinary spectral diagnostics are built from the relative positions of distinct energy levels. In a BPS sector at fixed charges, however, supersymmetry fixes the energy through the central-charge bound. The protected states then form a degenerate eigenspace, and the spacing distribution contains no information beyond the imposed degeneracy. This is a different obstruction from the failure of a finite-size random-matrix fit: the relevant spacings vanish identically before any statistical analysis is performed. The standard ETH constructions of Refs. [1, 2] and the spectral-chaos criterion of Ref. [20] therefore do not, by themselves, provide a diagnostic for this problem.

The natural object is the bundle of protected states over the space of couplings. Let $H(\lambda)$ be a family of Hamiltonians, let $P(\lambda)$ project onto a fixed-charge BPS eigenspace of rank D, and write $Q = 1 - P$. When the eigenspace remains isolated from the non-BPS spectrum, its off-fiber response is

$$X_\mu = Q\,\partial_\mu P\,P, \qquad (X_\mu)_{ma} = \frac{\langle m|\partial_\mu H|a\rangle}{E_{\mathrm{BPS}} - E_m}, \tag{40}$$

where $|a\rangle$ belongs to the protected fiber and $|m\rangle$ belongs to its complement. The connection on the fiber and its curvature are

$$(A_\mu)_{ab} = i\langle a|\partial_\mu b\rangle, \qquad F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu - i[A_\mu, A_\nu]. \tag{41}$$

Under a change of frame in the degenerate fiber, $F_{\mu\nu}$ transforms by conjugation. Its eigenvalues, traces of powers, and holonomies are therefore the appropriate gauge-covariant observables. The construction is the degenerate-state extension of the Berry phase and the Wilczek–Zee connection [6, 7]; it is also the object proposed specifically for BPS chaos in Ref. [21].

A. What is known about smooth and black-hole-motivated sectors

The distinction between smooth horizonless states and black-hole microstates is a statement about the geometry represented by a particular family of states, not a consequence of BPS protection alone. D-brane counting and the AdS/CFT correspondence provide the setting in which supersymmetric black-hole microstates can be compared with states represented by smooth geometries [22, 23]. They do not imply that every protected multiplet has the same response statistics.

The direct Berry-curvature comparison is supplied by Ref. [24]. For the studied 1/2-BPS states in the D1/D5 CFT, which are used there as smooth, horizonless examples, the curvature under marginal deformations is non-random and is often exactly zero at generic coupling. The qualification "often" is essential: the result concerns the sectors and deformations analyzed in that work, not every D1/D5 state or every possible tangent direction. The same source studies $\mathcal{N} = 4$ SYM and finds vanishing curvature for the analyzed 1/4-BPS sectors at generic coupling [24]. In both cases the conclusion is a statement about the matrix curvature of a degenerate bundle. It is not the claim that the corresponding Hamiltonian is integrable, nor that a vanishing curvature proves the absence of all forms of quantum chaos.

The contrast is with the paper's black-hole-motivated examples. In $\mathcal{N} = 2$ super-JT gravity, the Berry-curvature calculation gives random-matrix-like curvature statistics; in the $\mathcal{N} = 2$ SYK model, explicit numerical calculations give the same qualitative behavior [24]. The word "like" matters. The statement concerns the statistics of the curvature matrix after the specified deformation and symmetry prescription. It does not say that the entries form an independent Gaussian ensemble, that the distribution is Haar invariant, or that the result is insensitive to the choice of moduli coordinates. The gauge-invariant content is carried by conjugation-invariant functions of F, while the observable distribution may retain correlations between tangent directions.

This distinction is also visible in the relation between monotone and fortuitous BPS states. Supersymmetric SYK constructions and their protected sectors are described in Refs. [17, 25]; the finite-size appearance of additional relations is discussed in Ref. [26]. The label "fortuitous" is a statement about which BPS states exist at a given value of the number of degrees of freedom. It is not itself a proof of random curvature. Conversely, a

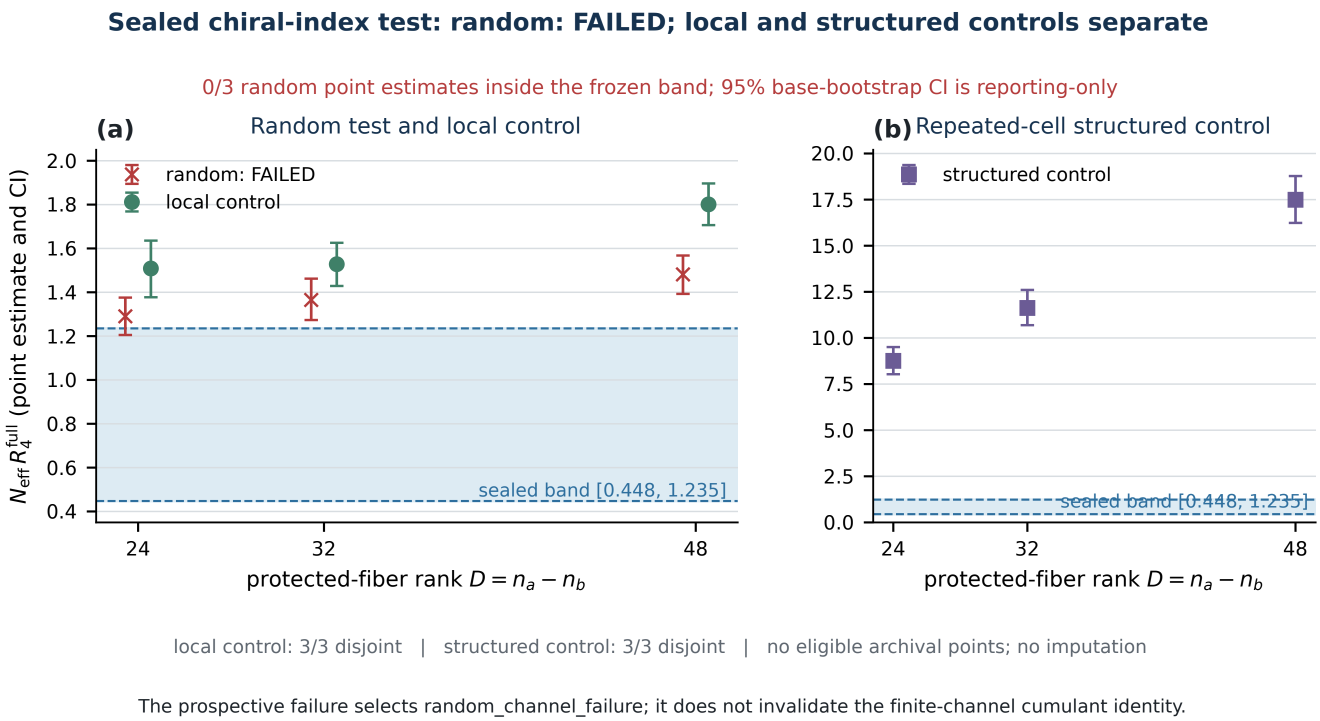

FIG. 4. Prospective chiral-index test. The shaded band is fixed from the three development sizes before validation. Markers and simultaneous intervals show means over 64 independent base matrices after averaging 16 tangents within each base. All three dense-random validation points lie above the sealed band, selecting `random_channel_failure`. Local and repeated-cell controls satisfy the separately frozen separation rule. Intervals report base-bootstrap uncertainty and do not alter the point-estimate decision.

monotone state is not defined by $F = 0$. The curvature comparison in Ref. [24] supplies the additional calculation needed to separate the two response behaviors.

B. The $\mathcal{N} = 2$ SYK and super-JT comparison

The SYK model is useful here because it supplies a controlled large-N setting with a well-developed low-energy description. The random spin-interaction precursor was introduced by Sachdev and Ye [27]. The modern SYK analysis identifies an emergent near-conformal regime and a reparametrization soft mode [25]; supersymmetric variants and their protected sectors are constructed in Ref. [17]. These facts establish why SYK is a useful model for testing a geometric diagnostic. They do not turn an SYK calculation into a theorem about higher-dimensional black holes.

The low-energy gravity comparison is similarly conditional. Nearly-AdS$_2$ dynamics is governed by a Schwarzian sector in the standard treatment [28], and the Schwarzian path integral and its correlators have controlled formulations [29, 30]. The direct statement relevant to this paper is narrower: the $\mathcal{N} = 2$ super-JT calculation in Ref. [24] produces random-matrix-like Berry curvature for the protected states considered there. Neither this calculation nor the SYK numerics establishes that all supersymmetric black-hole microstates have a universal curvature distribution.

The usual ETH language remains useful as a comparison of observables, not as a replacement for the geometric calculation. ETH concerns matrix elements of operators in energy eigenstates, whereas the present object is a connection on an exactly degenerate fiber. SYK ETH studies [31] and gravitational moment constructions [32] help relate the two settings, but neither source proves a Berry-curvature ETH statement. The difference is especially important at zero temperature, where a protected energy is fixed by a charge constraint while the state bundle may still have nontrivial holonomy.

C. A bounded relation to “It from ETH”

The phrase “It from ETH” refers here only to the construction of Ref. [33], which studies multi-interval entanglement and replica-wormhole structures in a large-c BCFT ensemble. That work is a relevant conceptual neighbor because it connects statistical information in a boundary ensemble to geometric data in a gravitational description. It does not calculate the non-Abelian Berry curvature of a BPS state bundle, and it does not validate the finite-channel law tested below. We therefore use the reference only to delimit a possible future bridge between state-bundle geometry, entanglement data, and replica constructions. No author attribution other than the official record is made here, and no unresolved name is inserted into the bibliography.

D. Claim boundary for the present manuscript

The direct evidence can be summarized without turning a model comparison into a universal black-hole statement:

1. the ordinary spectrum is silent inside an exactly degenerate BPS fiber;
2. the cited D1/D5 and $\mathcal{N} = 4$ SYM calculations provide non-random or vanishing curvature in the stated smooth/horizonless sectors;
3. the cited $\mathcal{N} = 2$ SYK and super-JT calculations provide random-matrix-like curvature in the stated black-hole-motivated models;
4. the $\mathcal{N} = 2$ SYK moduli-space calculation provides quantized Chern data, including exponentially large first Chern numbers in specified large-N sectors [24]; and
5. these results motivate tests of geometric response in additional protected systems, but do not establish that black holes satisfy an asymptotic or model-independent geometric-response law, ordinary ETH, or a thermalization law.

This boundary is the one used in the remainder of the paper. In particular, the black-hole language identifies a falsifiable target class of models; it is not used as an assumption that supplies the missing statistical closure.

VIII. TOPOLOGY OF THE PROTECTED-STATE BUNDLE

Local curvature statistics do not exhaust the information carried by a degenerate state bundle. The bundle can be locally close to a random matrix field and still obey global quantization conditions imposed by the topology of the moduli space. This distinction is central to the geometric formulation: local eigenvalue repulsion is a statement about a matrix at one point, whereas a Chern number is an integral constraint on a family of matrices over a closed cycle. The two statements can coexist, but neither implies the other.

A. The bundle and its gauge-covariant curvature

Let $\mathcal{M}_{\rm reg}$ denote a connected region of coupling space on which the protected rank is constant and the external gap remains open. The BPS eigenspaces form a rank-D complex vector bundle

$$\mathcal{E}_{\rm BPS} \longrightarrow \mathcal{M}_{\rm reg}, \qquad \mathcal{E}_{\rm BPS}|_\lambda = \operatorname{im} P(\lambda). \tag{42}$$

Choose a local orthonormal frame $|a(\lambda)\rangle$ for the fiber. The Berry connection and curvature are

$$(A_\mu)_{ab} = i\langle a|\partial_\mu b\rangle, \qquad F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu - i[A_\mu, A_\nu]. \tag{43}$$

Under a frame change $|a\rangle \mapsto |b\rangle U_{ba}(\lambda)$, one has $A \mapsto U^{-1}AU + iU^{-1}dU$ and $F \mapsto U^{-1}FU$. Consequently, $\operatorname{Tr} F$ and the eigenvalues of F are frame independent, although the individual matrix entries are not. This is the non-Abelian extension of the adiabatic phase and the geometric metric construction [6, 7, 9]. Kato's projector transport gives the corresponding formulation without choosing a frame [8].

The topology is meaningful only after the domain has been specified. If a submanifold of $\mathcal{M}$ contains a point where an additional BPS state enters the protected sector, or where the external gap closes, the rank-D bundle is not defined across that point. One must excise the singular locus and work on $\mathcal{M}_{\rm reg}$. Treating the excised locus as an ordinary point would confuse a change of bundle rank with a large but finite curvature.

B. Chern data as an integral constraint

For a closed oriented two-cycle $\Sigma \subset \mathcal{M}_{\rm reg}$, the first Chern number of the determinant line bundle is

$$c_1(\mathcal{E}_{\rm BPS}; \Sigma) = \frac{1}{2\pi} \int_\Sigma \operatorname{Tr} F \in \mathbb{Z}, \tag{44}$$

with the normalization in Eq. (43). The many-body formulation of quantized Hall response established that the same type of integral can be defined over a parameter torus even when the fiber is an interacting many-body ground-state sector [11]. On a discretized parameter space, link variables provide a gauge-invariant numerical implementation of the corresponding integer [34].

Equation (44) is stronger than a statement about the mean or variance of a local curvature entry. Small perturbations of the curvature that preserve the bundle and the cycle cannot change the integer. Conversely, a change in the integer requires a singularity, a change of cycle, or a loss of the rank-protection and gap assumptions. This is why a random matrix comparison must be made together with a topology audit: a genuinely independent field of matrices would not, without additional constraints, reproduce the quantized flux of a nontrivial bundle.

For a flat connection on a contractible patch, $F = 0$ implies that the local first Chern integral vanishes. This local implication should not be promoted to a statement that the entire moduli space is topologically trivial. A flat connection may have nontrivial holonomy on a non-simply-connected domain, and the domain itself may acquire nontrivial cycles after singular strata are removed.

(a) **Source-audited geometric response in protected gravitational sectors**

System / sector	Evidence label	Calculated geometric result	Claim boundary	Source
D1/D5 1/2-BPS horizonless	analytic calculation	Non-random and often vanishing Berry curvature in the studied marginal-deformation sectors.	Not every D1/D5 BPS state; no statistical-closure theorem.	[chen2026berry]
N=4 SYM 1/4-BPS	analytic calculation	Vanishing curvature at generic coupling for the sectors analyzed.	Sector-specific calculation, not all protected operators.	[chen2026berry]
N=2 super-JT	analytic calculation	Random-matrix-like Berry curvature in the super-JT model.	Does not extend to every supersymmetric gravity theory.	[chen2026berry]
N=2 SYK	numerical calculation	Random-matrix-like curvature; first Chern numbers grow exponentially in the stated sectors.	Finite-N checks through N=10; model and deformation dependent.	[chen2026berry]
broader black-hole interpretation	conjecture	Protected sectors may retain chaotic geometric signatures when level statistics are silent.	Motivation only; the v14 prospective channel-closure gate failed.	[chen2024bps; chen2026berry]

Evidence labels distinguish direct calculations from the broader conjecture. All citation keys are verified in the Paper II audit.

FIG. 5. Source-bounded comparison of protected sectors. Each row is generated from the primary-source audit and labels whether the cited work contains a calculation, numerical observation, or conjectural application. Smooth/horizonless and black-hole-motivated entries are not treated as two samples from a common ensemble. The comparison motivates response observables but supplies no title-gate evidence for the chiral channel law.

C. Topology in the $\mathcal{N} = 2$ SYK moduli space

The direct topological calculation relevant here is the $\mathcal{N} = 2$ SYK analysis of Ref. [24]. The paper identifies special loci in the coupling space where additional BPS states appear, studies the regular moduli space obtained after their removal, and integrates the Berry curvature over candidate two-spheres. It reports quantized Chern numbers and finds that the generic-point curvature matrix can resemble a random matrix while its integral remains constrained by Eq. (44).

For the family of two-spheres and charge sectors analyzed there, the reported first Chern number is

$$c_1^{(r)} = \frac{1}{2\pi}\int_\Sigma \mathrm{Tr}\, F^{(r)} = \binom{N}{r+3} - \binom{N}{r-3}. \qquad (45)$$

The formula is an empirical result of the stated moduli-space construction; the source reports checks through $N = 10$ and does not claim a systematic classification of all cycles. In particular, the even-N sector $r = N/2$ has vanishing value in the formula and must not be included in a blanket statement that every large-N sector carries an exponentially large integer.

For the neighboring even sector $r = N/2 - 1$, Eq. (45) gives the large-N estimate

$$c_1^{(N/2-1)} \sim 24\sqrt{\frac{2}{\pi}}\, N^{-3/2} 2^N, \qquad N \to \infty, \qquad (46)$$

as reported in Ref. [24]. The appropriate wording is thus that the studied $\mathcal{N} = 2$ SYK family contains exponentially large first Chern numbers in specified large-N sectors. It is not correct to state, without these qualifications, that all Chern numbers of $\mathcal{N} = 2$ SYK are exponential or that the asymptotic formula has been proved for every cycle.

The mechanism is transparent in the bundle language. The singular strata act as sources of Berry flux because the protected rank changes when they are approached. A two-sphere that avoids the singular strata can nevertheless enclose them in the ambient coupling space, giving a nonzero integral of $\mathrm{Tr}\, F$. The local curvature away from the strata can therefore show random-matrix-like eigenvalue statistics while the global flux remains an integer. This is a concrete form of structured geometric randomness: local randomness and global topological memory are not mutually exclusive [24].

D. Topology versus local random-matrix statistics

There are three logically distinct tests for a protected bundle:

1. a local test, such as eigenvalue repulsion or a distribution of curvature invariants at one point;

2. a transport test, such as a Wilson loop or a non-Abelian holonomy; and

3. a global test, such as a Chern number on a specified cycle.

The Jacobi and random-compression ensembles provide useful local reference models for the first test [35, 36]. They do not encode the topology of the parameter space. The Chern integral supplies an independent check that can reject a naive identification of the curvature field with an unconstrained collection of independent random matrices.

This point also limits the interpretation of non-Abelian curvature in the smooth sectors. If the direct calculation finds $F = 0$ for a particular family, then the first Chern integral over any cycle contained in that flat region is zero. The result does not exclude nontrivial topology elsewhere in the full moduli space, nor does it imply that a different BPS multiplet has a flat connection. The source-specific sector and deformation prescription must accompany every such statement [24].

Likewise, a random-matrix-like local spectrum in $\mathcal{N} = 2$ SYK or super-JT does not erase the Chern constraint. A geometric response ansatz that is intended to describe a full moduli space must preserve both properties: the local distribution of F and the integral quantization of its curvature. The next section formulates this as a conditional prediction rather than as a theorem about BPS black holes.

IX. CONDITIONAL PREDICTIONS AND FALSIFICATION CRITERIA

The theory and numerical gates impose a strict distinction between an exact finite-channel identity and a statistical closure hypothesis. For independent centered response channels Y_α in one common response space, with fixed weights w_α, the cumulants of

$$Z = \sum_\alpha w_\alpha Y_\alpha \tag{47}$$

obey the exact algebraic relation

$$\kappa_k(Z) = \sum_\alpha w_\alpha^k \kappa_k(Y_\alpha). \tag{48}$$

This identity follows from independence and does not require a large number of channels. It is therefore not the part of the proposal that is at risk in the numerical test. The additional claim is that, after complete complex covariance and pseudocovariance normalization, comparable channel cumulants produce a fourth-order scale controlled by

$$N_{\text{eff}} = \frac{\left(\sum_\alpha w_\alpha^2\right)^2}{\sum_\alpha w_\alpha^4}, \qquad \frac{\kappa_4(Z)}{\left(\sum_\alpha w_\alpha^2\right)^2} \propto N_{\text{eff}}^{-1}. \tag{49}$$

The proportionality in Eq. (49) is conditional on channel comparability, absence of a dominant weight, and an outcome-independent whitening map. It is not an exponent to be fitted after the data are seen.

The use of higher moments is motivated by the limitations of the leading ETH ansatz. Ordinary ETH concerns diagonal smoothness and off-diagonal matrix elements in an energy basis [1, 2]. Correlations needed for out-of-time-order observables and multi-operator products require additional information beyond a two-point covariance [37, 38]. The geometric version makes the same separation in a different space: it asks whether the tangent response of an exactly degenerate fiber has a closed distribution after its complete real-augmented covariance is fixed.

A. Status of the prospective chiral test

The chiral rectangular-kernel ensemble was the prospective test of this statistical closure. Its exact-index and projector-response checks passed, and the channel-whitening construction was well defined on all registered validation cases. Those algebraic checks establish that the numerical object was an isolated, exactly degenerate kernel response. They do not establish that its tangent channels are independent or that their fourth cumulants are comparable.

The sealed prediction was fixed from the three development sizes $(n_b, n_a) = (16, 24), (24, 36), (32, 48)$. The resulting simultaneous band for the covariance-whitened complete standardized fourth cumulant was

$$0.44763 \;\le\; N_{\text{eff}} R_4^{\text{full}} \;\le\; 1.23521. \tag{50}$$

The three sealed random-class validation point estimates at $(48, 72)$, $(64, 96)$, and $(96, 144)$ were, respectively,

$$1.29034, \qquad 1.36496, \qquad 1.48145. \tag{51}$$

Thus the primary random-channel prediction failed. The bootstrap interval for the first validation size overlaps the upper edge of the development band, whereas the

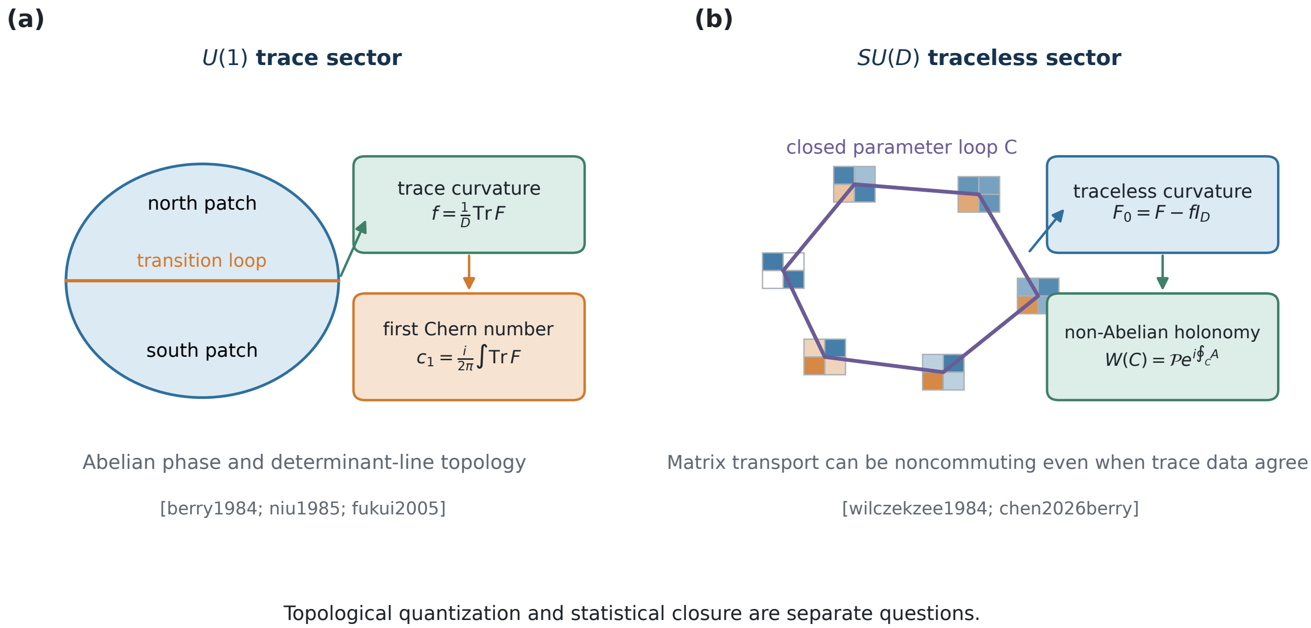

FIG. 6. Topology and non-Abelian statistics constrain different parts of the bundle. The trace curvature fixes the determinant-line $U(1)$ Chern class, whereas traceless $SU(D)$ curvature controls internal holonomy and connected matrix statistics. A fixed Chern number neither enforces nor excludes random non-Abelian geometry; topology is a global integral constraint, while statistical closure concerns the distribution of local matrix-valued response.

intervals at the two larger sizes are displaced above it; the registered rule nevertheless evaluates the primary prediction using the fixed point estimates and does not widen the band after unsealing. The local and structured controls separate from the random development band, but this does not rescue the failed random-channel branch. The resulting theory gate is therefore false.

This failure has a precise interpretation. It is evidence against the particular finite-size closure hypothesis, with its selected tangent ensemble, whitening prescription, and sealed statistic. It is not a disproof of the algebraic cumulant identity in Eq. (48). Nor does it identify the source of the discrepancy: correlated channels, noncomparable single-channel cumulants, finite-size geometry, or residual multivariate covariance and pseudocovariance tensor structure not captured by the scalar pooled statistic can all produce the observed departure. A failed closure test must not be relabeled as evidence for a different universal law.

The result also fixes the manuscript's title and conclusion boundary. The positive title and theorem-level conclusion "The Geometric ETH" are not licensed by the present gate. The bounded result is a conditional geometric response theory for exactly degenerate state bundles, together with a failed prospective random-channel validation. The distinction is substantive: a finite exact identity can remain useful even when the proposed ensemble does not obey its expected fourth-order scaling.

B. Predictions that remain testable

The failed gate does not leave the theory without predictions. It changes them from universal assertions to conditional tests with explicit failure modes.

a. Gauge-covariant response. For any protected rank-D bundle, a change of fiber frame sends $F_{\mu\nu} \mapsto U^{-1}F_{\mu\nu}U$. A valid numerical comparison must therefore be formulated in terms of eigenvalues, traces of powers, singular values, holonomies, or complete tensors transformed covariantly. A change in the histogram of individual matrix entries under a frame rotation is not a physical effect. This prediction is the direct geometric content of the Berry/Wilczek–Zee construction [6, 7] and of the BPS proposal [24].

b. Conditional inverse-participation scaling. If a new protected model supplies independently indexed channels, an outcome-independent channel map, finite fourth moments, and a non-dominant weight profile, then Eq. (49) predicts that the complete standardized fourth cumulant is controlled by $\sum_\alpha w_\alpha^4$. The test must freeze the covariance and pseudocovariance prescription before the validation data are opened. A failure should be reported as a failure of that channel class, not absorbed by refitting an exponent or changing the effective-channel definition.

c. Topology-constrained local statistics. In a model with a constant-rank protected bundle and noncontractible cycles, the local curvature law and the Chern integral must be tested together. The $\mathcal{N} = 2$ SYK calculation provides a concrete target: it reports random-matrix-like local curvature and, in specified sectors, exponentially large first Chern numbers [24]. A future ensemble that matches the local spectrum but violates integer flux is not a valid model of the same bundle. Conversely, integer flux alone does not imply random local statistics.

d. Protected-sector comparison. For a matched marginal-deformation prescription, the cited D1/D5 and $\mathcal{N}=4$ SYM sectors provide a non-random/flat comparison, while the cited $\mathcal{N}=2$ SYK and super-JT calculations provide a random-matrix-like comparison [24]. A new calculation should hold fixed the definition of the protected fiber, the external-gap criterion, the tangent normalization, and the gauge-invariant observable before comparing these classes. It must not infer a black-hole label solely from a finite histogram.

e. A conditional black-hole test. Suppose a future supersymmetric black-hole construction supplies an exactly degenerate BPS bundle, a controlled moduli-space deformation, and a positive external gap over the sampled region. Then one can test whether its curvature invariants follow the random-matrix-like behavior found in the model studies of $\mathcal{N}=2$ SYK and super-JT [24]. The prediction is conditional on those three inputs. It does not assert that all black holes have such a bundle, that their curvature is universal, or that curvature statistics imply thermalization.

C. A preregistered falsification protocol

The following checks are sufficient to falsify the conditional geometric response ansatz for a specified model without conflating distinct hypotheses:

1. verify the exact index, constant protected rank, and positive external gap on every sampled base point;
2. freeze the tangent ensemble, channel identification, covariance, pseudocovariance, and normalization on a development sample;
3. define the complete real-augmented fourth cumulant, including all complex Wick pairings, before the validation sample is generated;
4. use independent Hamiltonian or disorder realizations whenever the claim concerns an ensemble rather than tangent directions at one fixed Hamiltonian;
5. compare the held-out statistic with the sealed prediction without fitting a size exponent or widening the acceptance band; and
6. test a structured control with the same fiber rank, tangent count, and normalization, so that a failure can be distinguished from a generic estimator defect.

The chiral v14 study satisfies the first and the sealing requirements but fails the random primary prediction. It therefore supplies a falsification record, not a positive Geometric ETH validation. The next useful calculation is not a larger claim; it is a mechanism-resolved replication that changes one source of structure at a time and records whether the random-channel failure follows the response algebra, the tangent ensemble, or the topology of the bundle.

X. DISCUSSION AND LIMITATIONS

The calculation developed here separates a mathematical statement from a statistical one. A degenerate eigenspace defines a vector bundle over a coupling manifold, and its projector has a gauge-covariant response. Under independent centered channels, weighted cumulants add exactly. Neither fact requires a random-matrix assumption. The additional assertion that the complete fourth cumulant closes at an inverse effective-channel scale was tested in a sealed prospective chiral ensemble and failed its random primary branch. The appropriate conclusion is therefore a bounded geometric-response theory, not an asymptotic or model-independent theorem for protected-state ensembles.

A. What the present work establishes

The first result is a change of diagnostic object. In ordinary ETH, one studies matrix elements between energy eigenstates and their smooth dependence on energy, with the random component suppressed according to the Hilbert-space size [1, 2]. Spectral chaos supplies a related but distinct test through level fluctuations [20, 39]. Both constructions require nontrivial spectral variation. For a protected rank-D fiber at one BPS energy, the repeated eigenvalue is fixed by the symmetry constraint. The information that remains is how $P(\lambda)$ is embedded in the full Hilbert space and how it is transported around a loop.

The second result is the gauge-covariant formulation of that information. The off-fiber response $X_\mu = Q(\partial_\mu P)P$ and the non-Abelian curvature $F_{\mu\nu}$ are invariantly defined up to the natural action of the fiber frame group. This extends the Abelian geometric phase [6] to degenerate multiplets through the Wilczek–Zee connection [7], while Kato's projector transport supplies a frame-free description [8]. A nonzero response is therefore compatible with exact energy degeneracy; it does not require splitting the protected level.

The third result is a precise conditional channel statement. The cumulant identity is exact for independent centered channels, while the N_{eff}^{-1} scaling requires comparable standardized channel cumulants and a fixed complete covariance prescription. The response-algebra commutant can diagnose unresolved invariant blocks, but a scalar commutant does not imply a Gaussian field, ETH, or thermalization. This distinction prevents an algebraic irreducibility check from being mistaken for statistical closure.

The fourth result is negative but informative. The chiral-index parent has an exact kernel and passes the index, gap, response-equation, and sealing checks. Its random-channel validation nevertheless falls outside the frozen development band. Because the random branch was the primary prospective test, the theory gate is false. The local and structured controls separate from the random band, so the outcome is not explained by an across-the-board failure of the numerical construction. It remains ambiguous whether the failure comes from channel correlations, noncomparable channel cumulants, finite-rank geometry, or an incomplete model of the tangent ensemble. That ambiguity is a result to be resolved, not a license to reinterpret the outcome as a positive universal law.

B. Relation to the BPS and black-hole literature

The BPS application has a clear evidentiary boundary. The direct source [24] finds non-random and often vanishing Berry curvature for the studied 1/2-BPS D1/D5 horizonless sectors, and vanishing curvature for the studied 1/4-BPS $\mathcal{N} = 4$ SYM sectors at generic coupling. The same source reports random-matrix-like Berry curvature in its $\mathcal{N} = 2$ super-JT calculation and in explicit $\mathcal{N} = 2$ SYK numerics. It also reports exponentially large first Chern numbers in specified large-N sectors of the $\mathcal{N} = 2$ SYK moduli space. These are direct results of that reference, with the model, deformation, sector, and finite-N qualifications stated in Secs. VII and VIII; they are not additional numerical results obtained by the present chiral test.

The distinction between evidence and interpretation matters because the black-hole label carries physical assumptions. D-brane microstate counting and holographic duality establish important examples and the setting for comparing BPS states with black-hole geometries [22, 23]. They do not imply that every supersymmetric black-hole sector has a constant-rank bundle over a useful moduli space, a positive external gap throughout that space, or a universal curvature ensemble. Those are precisely the hypotheses a future calculation would have to check.

The SYK and JT references sharpen rather than remove this boundary. SYK has a controlled large-N conformal regime and a soft-mode description [17, 25]; nearly-AdS$_2$ and Schwarzian analyses provide the corresponding low-energy gravitational language [28–30]. Random-matrix behavior of conventional black-hole spectral observables is also well studied in matrix-integral and gravity settings [40, 41]. None of these results, alone, proves a universal Berry-curvature distribution for supersymmetric black holes. The direct Berry-curvature evidence remains the source-specific calculation of Ref. [24].

C. Why local randomness does not erase topology

The topology calculation supplies the sharpest reason to avoid a purely local random-matrix interpretation. On a closed two-cycle in the regular moduli space, $\int \mathrm{Tr}\, F/(2\pi)$ is an integer. The many-body Hall conductance construction and its lattice implementation make clear that this integer is a global property of a parameter-dependent state bundle [11, 34]. In the $\mathcal{N} = 2$ SYK example, the source reports local random-matrix-like curvature together with quantized Chern data and exponentially large values in selected sectors [24]. The global constraint is therefore a form of memory that is invisible to a one-point eigenvalue statistic.

This observation also changes how a geometric ETH ansatz should be formulated if it is revisited. A candidate field ensemble must specify both the local law of F and the allowed bundle topology. A random compression or Jacobi ensemble can be a local null model for relative subspace orientation [35, 36], but it is not a model of a full moduli-space bundle until its transition functions, singular loci, and Chern constraints are defined. A local fit that ignores these data can succeed while describing the wrong global object.

D. Relation to OTOCs, subsystem ETH, and "It from ETH"

The geometric response should not be advertised as a replacement for every chaos diagnostic. OTOCs probe operator growth and dynamical commutators; generalized ETH analyses emphasize the higher matrix-element correlations needed to reproduce them [37]. Subsystem ETH instead formulates thermalization in terms of reduced density matrices [42, 43]. Parametric eigenstate deformation provides a closer precedent because it treats motion in coupling space as a chaos-sensitive observable [5]. Even there, the observable and the limiting procedure differ from a non-Abelian curvature of an exactly degenerate fiber.

The cited "It from ETH" construction connects a large-c BCFT ensemble to multi-interval entanglement and replica-wormhole structures [33]. It is conceptually compatible with asking how statistical information can be encoded in geometric data, but it is not a Berry-curvature calculation and does not validate the chiral channel law. In this manuscript it is therefore a bounded neighboring direction, not evidence for a theorem.

E. What would be required for a stronger claim

Several missing ingredients would have to be supplied before the phrase "Geometric ETH" could denote an asymptotic or universal statement:

1. a single protection mechanism with a sequence of

growing protected ranks, rather than a collection of sequences whose algebra and tangent prescription change together;

2. a fixed operator or deformation class, stated in physical units, with a controlled positive external gap and a documented treatment of symmetry sectors;

3. independent Hamiltonian or disorder realizations whenever the claim concerns an ensemble, instead of treating many tangent directions at one base Hamiltonian as independent samples;

4. complete real-augmented covariance and pseudo-covariance estimation, including the uncertainty associated with the actual independent sampling unit;

5. local-statistics convergence, a topology audit of the regular moduli space, and a preregistered scaling observable; and

6. a positive validation against a frozen prediction in which the threshold, channel map, and normalization are not refit after the result is opened.

The present chiral failure shows why each condition matters. Increasing the ambient Hilbert-space dimension alone does not establish an effective-channel limit, and a successful control separation cannot compensate for failure of the primary random branch.

A future black-hole calculation would additionally need to state what plays the role of the coupling manifold, how the BPS fiber is selected at fixed charges, which singular loci are removed, and which observable is compared to the model results. Only after these data are fixed could one ask whether the random-matrix-like curvature found in the studied $\mathcal{N} = 2$ SYK and super-JT models survives in a new supersymmetric black-hole construction [24]. The appropriate output of such a study could be a conditional test, a counterexample, or a mechanism-dependent classification; none is assumed in advance.

F. Conclusion

Exactly degenerate energies do not make state motion unobservable. The projector, its off-fiber response, its non-Abelian curvature, and its global holonomy remain well-defined whenever the protected rank and external gap are controlled. This supplies a meaningful geometric question in a regime where ordinary level statistics are silent.

The present paper establishes the formal response and the exact finite-channel cumulant identity under stated independence assumptions. It does not establish the proposed random-channel closure: the sealed chiral validation failed, so the theory gate is false. It therefore does not establish an asymptotic, universal, thermalizing, or black-hole Geometric ETH, and the positive title "The Geometric ETH" is not justified by the current evidence.

The defensible conclusion is narrower. Non-Abelian quantum geometry is a well-defined probe of how an exactly degenerate protected subspace moves, and the literature contains a source-specific contrast between non-random or flat curvature in the studied smooth/horizonless BPS sectors and random-matrix-like curvature in the studied $\mathcal{N} = 2$ SYK and super-JT models [24]. Turning that contrast into a general law remains an open, falsifiable problem whose next step is a mechanism-controlled validation with independent sampling and topology preserved.

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